

RW3 Separate Sheet Module - ConsumerUA

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Input workbook: full_uah.xlsx

Source sheet: Consumer_UA

Rows (clean): 5463

Date range: 2021-02-01 .. 2026-01-26

Products: 3

Stats rows: 9

Tests rows: 9

Interpretation option: use tests table -> recommended_action/recommended_model_family per product.

Table: clean

source	date	raw	product	standardized	type	brand	region	price	price_real_input	price_linear_input	price_pchip_input	unit_ok	
ConsumerUA	2021-02-01	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.422818971896502	nan	6.422818971896502	6.422818971896502	1
ConsumerUA	2021-02-02	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.417271543787608	nan	6.417271543787608	6.410999151557978	1
ConsumerUA	2021-02-03	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.411724115678714	nan	6.411724115678714	6.401691643587824	1
ConsumerUA	2021-02-04	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.40617668756982	nan	6.40617668756982	6.394643539010946	1
ConsumerUA	2021-02-05	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.400629259460927	nan	6.400629259460927	6.389601928852245	1
ConsumerUA	2021-02-06	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.395081831352033	nan	6.395081831352033	6.386313904136625	1
ConsumerUA	2021-02-07	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.389534403243139	nan	6.389534403243139	6.38452655588899	1
ConsumerUA	2021-02-08	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.383986975134245	nan	6.383986975134245	6.383986975134245	1
ConsumerUA	2021-02-09	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.393665249836379	nan	6.393665249836379	6.386226115189854	1
ConsumerUA	2021-02-10	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.403343524538513	nan	6.403343524538513	6.392372515767428	1
ConsumerUA	2021-02-11	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.413021799240647	nan	6.413021799240647	6.401569647483095	1
ConsumerUA	2021-02-12	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.422700073942781	nan	6.422700073942781	6.41296098095298	1
ConsumerUA	2021-02-13	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.432378348644915	nan	6.432378348644915	6.425689986793206	1
ConsumerUA	2021-02-14	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.442056623347049	nan	6.442056623347049	6.438900135619899	1
ConsumerUA	2021-02-15	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.451734898049183	nan	6.451734898049183	6.451734898049183	1
ConsumerUA	2021-02-16	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.468838363316452	nan	6.468838363316452	6.467448816003723	1
ConsumerUA	2021-02-17	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.485941828583722	nan	6.485941828583722	6.488084155395553	1
ConsumerUA	2021-02-18	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.503045293850992	nan	6.503045293850992	6.510965972824841	1
ConsumerUA	2021-02-19	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.520148759118261	nan	6.520148759118261	6.533419324891764	1
ConsumerUA	2021-02-20	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.53725222438553	nan	6.53725222438553	6.552769268196494	1
ConsumerUA	2021-02-21	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.5543556896528	nan	6.5543556896528	6.566340859339205	1
ConsumerUA	2021-02-22	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.57145915492007	nan	6.57145915492007	6.57145915492007	1
ConsumerUA	2021-02-23	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.373633999016056	nan	6.373633999016056	6.494751441406269	1
ConsumerUA	2021-02-24	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	6.175808843112041	nan	6.175808843112041	6.296926285502254	1
ConsumerUA	2021-02-25	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.977983687208027	nan	5.977983687208027	6.026430664164113	1
ConsumerUA	2021-02-26	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.780158531304012	nan	5.780158531304012	5.731711554347927	1
ConsumerUA	2021-02-27	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.582333375399998	nan	5.582333375399998	5.461215933009785	1
ConsumerUA	2021-02-28	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.384508219495983	nan	5.384508219495983	5.263390777105771	1
ConsumerUA	2021-03-01	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.186683063591969	nan	5.186683063591969	5.186683063591969	1
ConsumerUA	2021-03-02	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.206968786523972	nan	5.206968786523972	5.19454895615744	1
ConsumerUA	2021-03-03	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.227254509455975	nan	5.227254509455975	5.214834679089442	1
ConsumerUA	2021-03-04	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.247540232387977	nan	5.247540232387977	5.242572300241364	1
ConsumerUA	2021-03-05	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.267825955319979	nan	5.267825955319979	5.2727938874466593	1
ConsumerUA	2021-03-06	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.288111678251981	nan	5.288111678251981	5.300531508618514	1
ConsumerUA	2021-03-07	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.308397401183984	nan	5.308397401183984	5.320817231550516	1
ConsumerUA	2021-03-08	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.328683124115987	nan	5.328683124115987	5.328683124115987	1
ConsumerUA	2021-03-09	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.312633630663558	nan	5.312633630663558	5.324088569603367	1
ConsumerUA	2021-03-10	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.296584137211128	nan	5.296584137211128	5.311839419221308	1
ConsumerUA	2021-03-11	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.280534643758699	nan	5.280534643758699	5.294237442703512	1
ConsumerUA	2021-03-12	Молоко	пастеризоване	жирн...	Молоко	питне	milk	Україна	5.26448515030627	nan	5.26448515030627	5.27358440978368	1

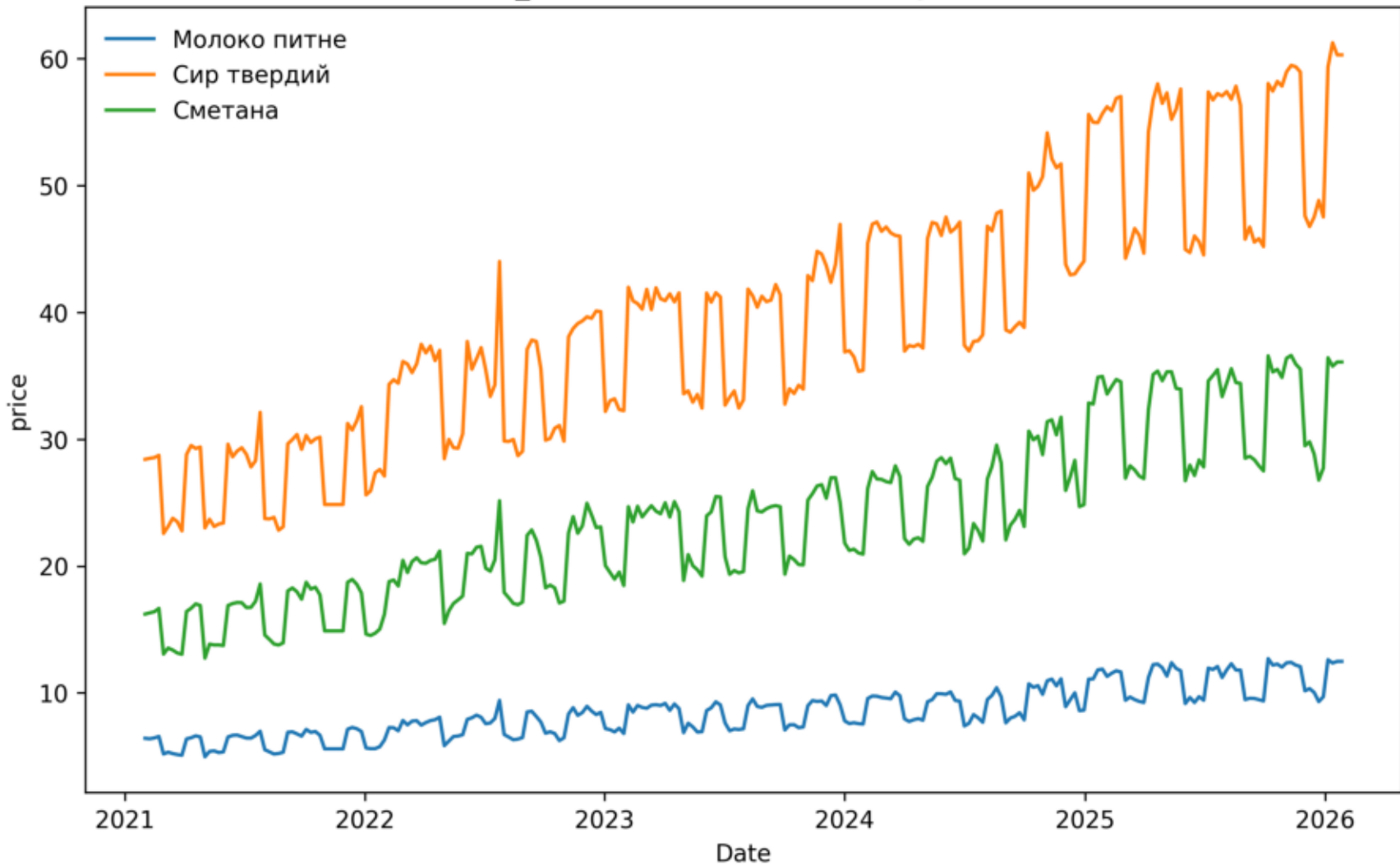
Table: descriptive_stats

source	product	standardized_type	series_variant	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
ConsumerUA	Молоко питне	milk	linear	1821	0	0.0	8.488818804549318	8.332297473913943	1.9183244717310495	4.965919408690954	12.70165401761612	5.518558193237996	6.989415113869986	9.623338467919126	12.09958365353656	0.3220977052607414	-0.6643326584169502	0.22598249720007962	0.006156032380013168	0.012370294913103989
ConsumerUA	Молоко питне	milk	pchip	1821	0	0.0	8.48878989674149	8.329208157414444	1.9278758951600201	4.965919408690954	12.70165401761612	5.459859817870655	6.962365468714518	9.625106769774382	12.12589662153451	0.3236920898041367	-0.6638762669593468	0.22710844756566012	0.00664321470961066	0.013286827995544943
ConsumerUA	Молоко питне	milk	real	1821	0	nan	8.488818804549318	8.332297473913943	1.9183244717310495	4.965919408690954	12.70165401761612	5.518558193237996	6.989415113869986	9.623338467919126	12.09958365353656	0.3220977052607414	-0.6643326584169502	0.22598249720007962	0.006156032380013168	0.012370294913103989
ConsumerUA	Сир твердий	hard_cheese	linear	1821	0	0.0	39.72249504221285	38.92611865995162	9.867994848211774	22.56905016755205	61.25074433539783	24.27669102755834	31.57378129715967	46.42297819341645	57.32643137729057	0.30179846741099375	-0.7573453144323103	0.24842333890973156	0.005816357202364242	0.012064634624935337
ConsumerUA	Сир твердий	hard_cheese	pchip	1821	0	0.0	39.72252358640556	38.89108735406067	9.900368709799574	22.56905016755205	61.25074433539783	23.83403391487175	31.611182080858	46.4024934881427	57.4033024664991	0.30396327496060677	-0.753668917388286	0.24923816051777303	0.0063479679422097715	0.013027872279663745
ConsumerUA	Сир твердий	hard_cheese	real	1821	0	nan	39.72249504221285	38.92611865995162	9.867994848211774	22.56905016755205	61.25074433539783	24.27669102755834	31.57378129715967	46.42297819341645	57.32643137729057	0.30179846741099375	-0.7573453144323103	0.24842333890973156	0.005816357202364242	0.012064634624935337
ConsumerUA	Сметана	sour_cream	linear	1821	0	0.0	23.599633115009688	23.08745964093858	6.22029124227194	12.73996201071674	36.60372619440565	14.46976675130172	18.62440056135446	27.60791055146646	35.21744832682492	0.37246705260679436	-0.7323701079718203	0.26357576035009417	0.005985686288159704	0.012136444771573801
ConsumerUA	Сметана	sour_cream	pchip	1821	0	0.0	23.59966646930827	23.08865290397035	6.2417445136830665	12.73996201071674	36.60372619440565	14.34276263484893	18.585584435018	27.61162012159911	35.27421492552332	0.3741017672987549	-0.7303024733987469	0.26448443760002077	0.00648308209561429	0.01305362430794103
ConsumerUA	Сметана	sour_cream	real	1821	0	nan	23.599633115009688	23.08745964093858	6.22029124227194	12.73996201071674	36.60372619440565	14.46976675130172	18.62440056135446	27.60791055146646	35.21744832682492	0.37246705260679436	-0.7323701079718203	0.26357576035009417	0.005985686288159704	0.012136444771573801

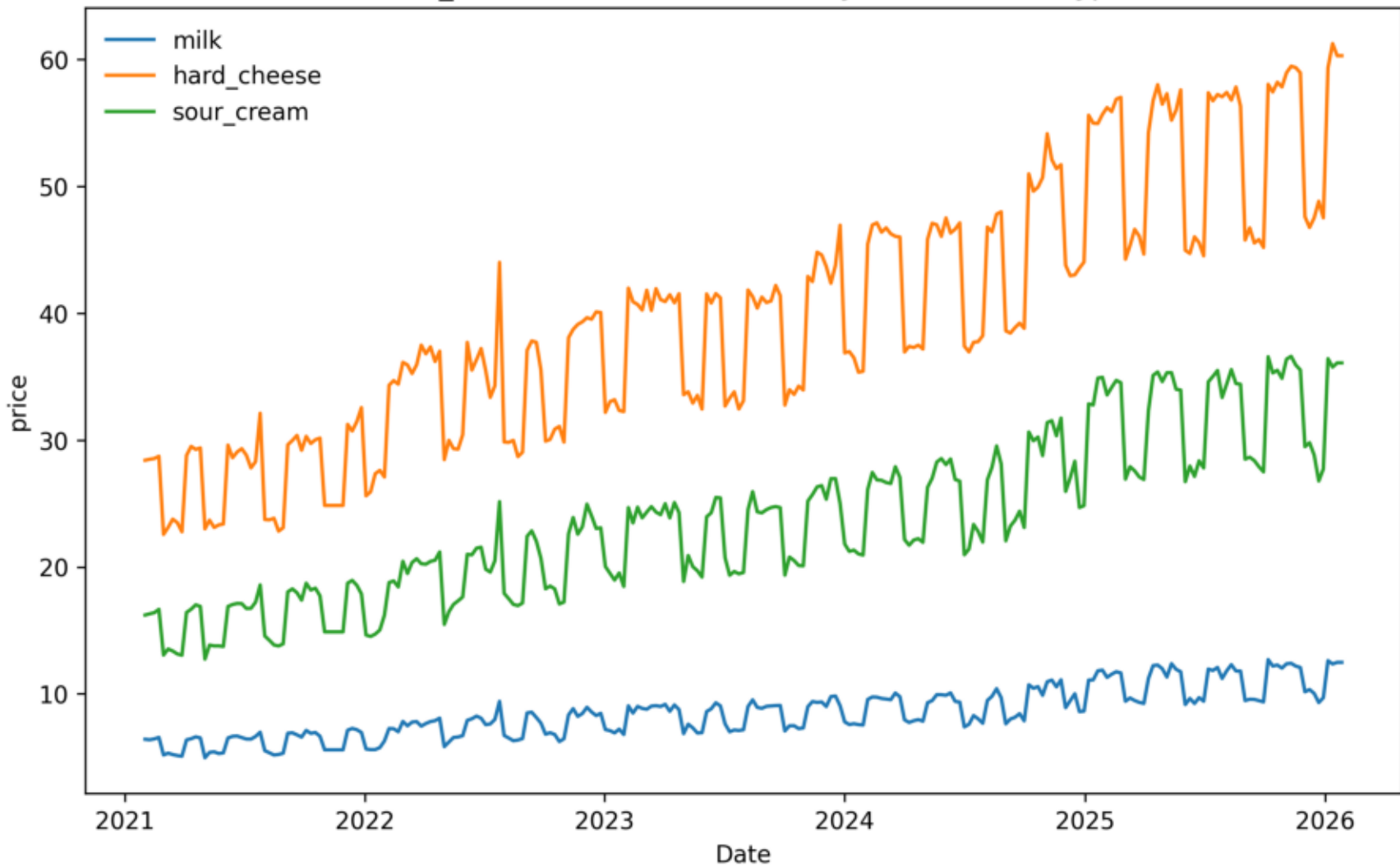
Table: tests

source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note
ConsumerUA	Молоко питне	milk	linear	261	0.9132067870531055	0.01	1.0726043916799254e-24	0.6325824362013026	0.0002906002997037569	2.896175101960723e-08	1	2.780434757935616	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Молоко питне	milk	pchip	261	0.908751351808557	0.01	1.5937754434907405e-22	0.6357350589466056	0.0007191266858439764	6.107916132817325e-06	1	2.7804926053180976	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Молоко питне	milk	real	261	0.9132067870531055	0.01	1.0726043916799254e-24	0.6325824362013026	0.0002906002997037569	2.896175101960723e-08	1	2.780434757935616	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сир твердий	hard_cheese	linear	261	0.90720218068931	0.01	3.7296570751491604e-30	0.6500692222339446	0.002287293481431856	1.4161604384975736e-07	1	14.842197109522267	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сир твердий	hard_cheese	pchip	261	0.9089732197811701	0.01	3.563486238386321e-29	0.5716530589887602	0.0023553102133987485	1.8595928666559863e-06	1	14.8421399897696	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сир твердий	hard_cheese	real	261	0.90720218068931	0.01	3.7296570751491604e-30	0.6500692222339446	0.002287293481431856	1.4161604384975736e-07	1	14.842197109522267	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сметана	sour_cream	linear	261	0.9251508147961878	0.01	1.0544712217033645e-26	0.5234286776283983	0.0008643386539634529	6.595261728597874e-09	1	9.384196037802763	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сметана	sour_cream	pchip	261	0.9208444482609317	0.01	8.300559229707679e-25	0.50291886584042	0.002340780855287725	1.674294341989109e-06	1	9.384129292552526	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...
ConsumerUA	Сметана	sour_cream	real	261	0.9251508147961878	0.01	1.0544712217033645e-26	0.5234286776283983	0.0008643386539634529	6.595261728597874e-09	1	9.384196037802763	non-stationary: differenc...	I(1)-like	high	high	high	high	Use differences or cointe...	ARDL/ECM/VECM (cointegrat...	OLS in levels without coi...	pattern consistent with p...

sheet_consumerua: Time Series by Product



sheet_consumerua: Time Series by Standardized Type



sheet_consumerua_distribution.png

sheet_consumerua: Distribution of price

